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(54) **METHOD FOR DETECTING CORONAVIRUS (SARS-CoV-2)**

# FOREIGN PATENT DOCUMENTS

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None  
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(57) **ABSTRACT**

Disclosed are oligonucleotide primers that hybridize specifically with any base sequence designed from the base sequences of the N gene, RNA-dependent RNA polymerase gene, M gene, and S gene of SARS-CoV-2, a nucleic acid amplification method using said primers, a test method for SARS-CoV-2 infection by detection of nucleic acid amplification, and a COVID-19 test kit.

**16 Claims, No Drawings**

**Specification includes a Sequence Listing.**

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**METHOD FOR DETECTING CORONAVIRUS  
(SARS-CoV-2)**

## RELATED PATENT APPLICATIONS

This application is based on and claims the benefit of priority from International Application No. PCT/JP2021/008261, filed on Mar. 3, 2021, which claims priority to Japanese Patent Application No. 2020-038195, filed on Mar. 5, 2020, and claims priority to Japanese Patent Application No. 2020-191630, filed on Nov. 18, 2020, the contents of which are hereby incorporated by reference in their entirety.

## SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Aug. 23, 2022, is named "051723-0571005\_SequenceListing.txt" and is 46.2 KB in size.

## TECHNICAL FIELD

The present invention relates to a method for detecting a new coronavirus (SARS-CoV-2) discovered in 2019, and more specifically to a method for assisting diagnosis of acute respiratory disease (COVID-19) due to the new 2019 coronavirus utilizing a highly sensitive method for detecting genes.

## BACKGROUND ART

Acute respiratory disease (COVID-19) due to the new 2019 coronavirus was reported in December 2019 as outbreaks of patients with pneumonia of unknown causes in Wuhan City, Hubei Province, China. The cause was confirmed to be a new coronavirus on Jan. 7, 2020, and the following week, infected persons were confirmed in a plurality of countries in addition to China. As of the end of January, the number of infected persons exceeded 10,000, but the spread of the infection did not stop, and the World Health Organization (WHO) declared a Public Health Emergency of International Concern (PHEIC) on January 31st.

The main clinical symptoms of COVID-19 are fever, cough, and general malaise. When it becomes severe, it causes pneumonia or dysfunction of the lungs, the heart, and the kidneys, which may lead to death. Although the mortality is about 3% which is lower than that of the SARS epidemic from 2002 to 2003, the number of infected persons is much higher than that of SARS. Therefore, the death toll is higher than that of the SARS epidemic.

The new virus (SARS-CoV-2) that causes COVID-19 is a betacoronavirus similar to the SARS coronavirus or MERS coronavirus. The genes of the new virus consist of a positive strand of single-stranded RNA and have a length of about 30,000 bases (GENBANK Accession No. MN908947).

The entire sequence of SARS-CoV-2 was determined and published about 1 month after the first report of the outbreak of pneumonia in patients. Based on the sequence, methods for detecting SARS-CoV-2 through RT-PCR have been developed (Non Patent Literature 1 and others).

However, since these detection methods were developed in a short period of time, some of them have problems in detection accuracy. For example, it has been announced that some of test kits developed by The Centers for Disease Control and Prevention (CDC) have defects (refer to Non

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Patent Literature 2). In addition, the number of tests required is also enormous due to the rapid spread of infection. For these reasons, a faster and more accurate method for detecting SARS-CoV-2 is required.

## CITATION LIST

## Non Patent Literature

- [Non Patent Literature 1] National Institute of Infectious Disease: Pathogen Detection Manual 2019-nCoV Ver.2.7 (edited on Feb. 25, 2020), Internet <URL: <https://www.niid.go.jp/niid/images/lab-manual/2019-nCoV20200225.pdf>>
- [Non Patent Literature 2] CDC Tests for COVID-19, retrieved on Feb. 18, 2020, Internet <URL: <https://www.cdc.gov/coronavirus/2019-ncov/about/testing.html>>

## SUMMARY OF INVENTION

## Technical Problem

An object of the present invention is to detect SARS-CoV-2, which is a pathogenic virus of COVID-19, with high sensitivity for diagnosis of COVID-19.

## Solution to Problem

The present inventors have conducted extensive studies to solve the above-described problem, and as a result, they have found that SARS-CoV-2 can be detected with high sensitivity by producing an oligonucleotide primer that is selectively hybridized with a sequence specific to SARS-CoV-2 and amplifying the sequence specific to SARS-CoV-2 through a LAMP method, thus leading to realization of the present invention.

That is, the present invention has the following configuration.

[1] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one designed from regions (a) and (b):

(a) a region from a 28976th base to a 29211th base of SEQ ID NO: 1; and

(b) a region from a 15394th base to a 15595th base of SEQ ID NO: 1.

[2] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one selected from the group consisting of (a1) and (b1):

(a1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 2, a B3 primer consisting of SEQ ID NO: 3, an FIP primer consisting of SEQ ID NO: 4, and a BIP primer consisting of SEQ ID NO: 5; and

(b1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 8, a B3 primer consisting of SEQ ID NO: 9, an FIP primer consisting of SEQ ID NO: 10, and a BIP primer consisting of SEQ ID NO: 11.

[3] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one selected from the group consisting of (a2) and (b2):

(a2) an LF primer consisting of SEQ ID NO: 6 and an LB primer consisting of SEQ ID NO: 7 in addition to the primer set of (a1) of [1]; and

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- (b2) an LF primer consisting of SEQ ID NO: 12 and an LB primer consisting of SEQ ID NO: 13 in addition to the primer set of (b1) of [1].
- [4] A method for detecting SARS-CoV-2 virus, comprising performing an amplification reaction through a LAMP method using the primer set of any one of [1] to [3].
- [5] A method for testing for COVID-19, comprising detecting amplification of a target nucleic acid region of SARS-CoV-2 virus using the primer set of any one of [1] to [3] to test for presence or absence of infection with SARS-CoV-2 virus.
- [6] A kit comprising the primer set of any one of [1] to [3] in a method for diagnosing COVID-19.
- [7] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one designed from regions (c) and (d):
- (c) a region from a 26767th base to a 26977th base of SEQ ID NO: 1; and
- (d) a region from a 24660th base to a 24916th base of SEQ ID NO: 1.
- [8] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one selected from the group consisting of (c1) and (d1):
- (c1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 14, a B3 primer consisting of SEQ ID NO: 15, an FIP primer consisting of SEQ ID NO: 16, and a BIP primer consisting of SEQ ID NO: 17; and
- (d1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 21, a B3 primer consisting of SEQ ID NO: 22, a B4 primer consisting of SEQ ID NO: 23, an FIP primer consisting of SEQ ID NO: 24, and a BIP primer consisting of SEQ ID NO: 25.
- [9] A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method, wherein the primer set consists of at least one selected from the group consisting of (c2) and (d2):
- (c2) a primer set consisting of an F3 primer consisting of SEQ ID NO: 14, a B3 primer consisting of SEQ ID NO: 15, an FIP primer consisting of SEQ ID NO: 16, a BIP primer consisting of SEQ ID NO: 17, an LF primer consisting of SEQ ID NO: 18, and an LB primer consisting of SEQ ID NO: 19; and
- (d2) a primer set consisting of an F3 primer consisting of SEQ ID NO: 21, a B3 primer consisting of SEQ ID NO: 22, a B4 primer consisting of SEQ ID NO: 23, an FIP primer consisting of SEQ ID NO: 24, a BIP primer consisting of SEQ ID NO: 25, an LF primer consisting of SEQ ID NO: 26, and an LB primer consisting of SEQ ID NO: 27.
- [10] A kit for detecting SARS-CoV-2 virus through amplification by a LAMP method, comprising the primer set of any one of [7] to [9]; and a fluorescence labeling probe.
- [11] The kit of [10], wherein the primer set is a primer set of (c1) or (c2), and the fluorescence labeling probe is a probe consisting of SEQ ID NO: 20.
- [12] The kit of [10], wherein the primer set is a primer set of (d1) or (d2), and the fluorescence labeling probe is a probe consisting of SEQ ID NO: 28.
- [13] The kit of any one of [10] to [12], which is used for testing for COVID-19.
- [14] A method for detecting SARS-CoV-2 virus, comprising performing an amplification reaction through a

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- LAMP method using the primer set of any one of [7] to [9] or the kit of any one of [10] to [13].
- [15] A method for testing for COVID-19, comprising detecting amplification of a target nucleic acid region of SARS-CoV-2 virus using the primer set of any one of [7] to [7] or the kit of any one of [10] to [13] to test for presence or absence of infection with SARS-CoV-2 virus.

## Advantageous Effects of Invention

According to the present invention, SARS-CoV-2 can be promptly detected with high sensitivity by producing an oligonucleotide primer that is selectively hybridized with a sequence specific to SARS-CoV-2 and amplifying the sequence specific to SARS-CoV-2 through a LAMP method.

## DESCRIPTION OF EMBODIMENTS

Examples of samples used in the present invention include specimens, such as sputum, bronchoalveolar lavage fluid, nasal mucus, nasal suction fluid, nasal lavage fluid, nasal swabs, pharyngeal swabs, mouthwash, saliva, blood, serum, plasma, cerebrospinal fluid, urine, feces, and tissue, derived from humans or other animals suspected of being infected with SARS-CoV-2 virus. In addition, cells used in infection experiments or the like or their culture solutions, or specimens derived from living bodies or specimens containing viruses isolated from cultured cells or the like can also be used as samples. These samples may be subjected to pretreatment such as separation, extraction, concentration, and purification.

Amplification of nucleic acids contained in a sample is achieved by a loop-mediated isothermal amplification method (WO 00/28082) called a LAMP method that is a nucleic acid amplification method which was developed by Notomi et al. and does not require temperature control indispensable for a PCR method. This method is a nucleic acid amplification method which enables an isothermal complementary strand synthesis reaction by combining primers which anneal their own 3' terminals to template nucleotides to serve as starting points for complementary strand synthesis and are annealed to loops formed at this time. In addition, in the LAMP method, since the 3' terminal of a primer is always annealed to a region derived from a sample, a check mechanism due to a complementary bond of sequences functions repeatedly. As a result, the LAMP method enables a nucleic acid amplification reaction with high sensitivity and high specificity.

Regarding oligonucleotide primers used in a LAMP method, there are at least 4 types of primers recognizing a sequence in a total of 6 regions of the sequence of a template nucleic acid, that is, regions F3c, F2c, and F1c from the 3' terminal side and regions B3, B2, and B1 from the 5' terminal side, which are called a forward inner primer, a backward inner primer, a forward outer primer, and a backward outer primer. In addition, complementary sequences of F1c, F2c, and F3c are respectively called F1, F2, and F3, and complementary strands of B1, B2, and B3 are respectively called B1c, B2c, and B3c. An inner primer is an oligonucleotide which recognizes a "specific nucleotide sequence region" on a target base sequence, has a sequence, providing a starting point for synthesis, at the 3' terminal, and simultaneously has a sequence complementary to an arbitrary region of a nucleic acid synthesis reaction product with this primer as a starting point, at the 5' terminal. Here, a primer containing a "sequence selected from F2" and

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a “sequence selected from F1c” is called a forward inner primer (hereinafter, an FIP primer), and a primer containing a “sequence selected from B2” and a “sequence selected from B1c” is called a backward inner primer (hereinafter, a BIP primer). On the other hand, an outer primer is an oligonucleotide which recognizes a “specific nucleotide sequence region” present on the 3' terminal side of a “specific nucleotide sequence region” on a target sequence and has a sequence providing a starting point for synthesis. Here, a primer containing a “sequence selected from F3” is called a forward outer primer (hereinafter, an F3 primer), and a primer containing a “sequence selected from B3” is called a backward outer primer (hereinafter, a B3 primer). Here, F in each primer represents a primer which complementarily binds to a sense strand of a target sequence and provides a starting point for synthesis, while B represents a primer which complementarily binds to an antisense strand of a target sequence and provides a starting point for synthesis. Here, the length of an oligonucleotide used as a primer is greater than or equal to 10 bases and preferably greater than or equal to 15 bases, and may be either chemically synthesized or natural. Each primer may be a single oligonucleotide or may be a mixture of a plurality of oligonucleotides.

In a LAMP method, in addition to the inner primers and the outer primers, another primer, that is, a loop primer, can be used. A loop primer is a primer having a sequence complementary to a sequence of a single-stranded portion of a loop structure on the 5' terminal side of a dumbbell structure. When this primer is used, the number of starting points for nucleic acid synthesis increases, whereby the reaction time can be reduced and the detection sensitivity can be increased (WO 02/024902). A sequence of a loop primer may be a sequence of a target gene or selected from a complementary strand thereof, or may be other sequences as long as it is complementary to the above-described sequence of a single-stranded portion of a loop structure on the 5' terminal side of a dumbbell structure. In addition, the loop primer may be one kind or two or more kinds, and is called a forward loop primer (hereinafter, LF) and a backward loop primer (hereinafter, LB) in the present specification.

In a case where a gene of a single-stranded RNA virus is detected through amplification by a LAMP method, an outer primer can be further added to increase the efficiency of a reverse transcription reaction in an initial stage of the amplification reaction. Since SARS-CoV-2 is a plus-strand single-stranded RNA virus, an outer primer (B4 primer) may be added to the downstream side of a B3 primer.

SARS-CoV-2 is an RNA virus. In a LAMP method, in a case where RNA is used as a template, a nucleic acid amplification reaction can similarly proceed by adding a reverse transcriptase to a reaction solution in a case where DNA is used as a template (RT-LAMP method).

The present inventors have conducted extensive studies on base sequences of primers in a LAMP method capable of promptly amplifying a sequence specific to SARS-CoV-2, and a combination thereof, and as a result, they have designed primer sets from an N gene region and an RNA-dependent RNA polymerase (RdRP) gene region of a sequence (SEQ ID NO: 1) of SARS-CoV-2. That is, the present invention is a primer set consisting of at least one designed from an N gene region (a region from a 28976th base to a 29211th base of SEQ ID NO: 1) and an RNA-dependent RNA polymerase (RdRP) gene region (a region from a 15394th base to a 15595th base of SEQ ID NO: 1) of a sequence (SEQ ID NO: 1) of SARS-CoV-2. The present

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invention is a primer set consisting of F3, B3, FIP, and BIP and designed as being a primer set comprising at least one selected from the following two sets a and b. In addition, the present invention is a primer set comprising an LF primer and an LB primer and comprising at least one selected from the following two sets a and b.

TABLE 1

|                       | Primer name | Base sequence                                   | SEQ ID NO. |
|-----------------------|-------------|-------------------------------------------------|------------|
| Primer set a (N gene) | F3          | TCTGGTAAAGGCCAAC                                | 2          |
|                       | B3          | GCTGAAGCGCTGG                                   | 3          |
|                       | FIP         | AGTACGTTTTTGCCGAGGC-AAGGCCAACTGTCTACT           | 4          |
|                       | BIP         | GTAACACAAGCTTTCGGCAG-ATTAGTTCCTGGTCCCCAA        | 5          |
|                       | LF          | AAGCCTCAGCAGCAGAT                               | 6          |
|                       | LB          | TCCAGAACAACCCAAGG                               | 7          |
| Primer set b (RdRP)   | F3          | TTCTATAGATTAGCTAATGAGTGT                        | 8          |
|                       | B3          | ACTTATCGGCAATTTGTGTACC                          | 9          |
|                       | FIP         | GAGGTTCCACCTGGTTTAACATAT-CTCAAGTATTGAGTGAAATGGT | 10         |
|                       | BIP         | CAGGAGATGCCACAACCTGCTTAT-AGATAAAAGTGCATTAACTTGG | 11         |
|                       | LF          | GAACCGCCACACATGACC                              | 12         |
|                       | LB          | CATTGTCAAGCTGTCCACGG                            | 13         |

In addition, the present inventors have conducted extensive studies on base sequences of primers in a LAMP method capable of promptly amplifying a base sequence specific to SARS-CoV-2, and a combination thereof, and as a result, they have designed primer sets from an M gene region and an S gene region of a sequence (SEQ ID NO: 1) of SARS-CoV-2. That is, the present invention is a primer set consisting of at least one designed from an M gene region (a region from a 26767th base to a 26977th base of SEQ ID NO: 1) and S gene region (a region from a 24660th base to a 24916th base of SEQ ID NO: 1) of a sequence (SEQ ID NO: 1) of SARS-CoV-2. The present invention is a primer set which consists of F3, B3 (or B3 and B4), FIP, and BIP and consists of at least one selected from the following two sets c and d. In addition, the present invention is a primer set comprising an LF primer and an LB primer and comprising at least one selected from the following two sets c and d.

TABLE 2

|                       | Primer name | Base sequence                                               |
|-----------------------|-------------|-------------------------------------------------------------|
| Primer set c (M gene) | F3          | TCGCAATGGCTTGCTCTTG (SEQ ID NO: 14)                         |
|                       | B3          | GCAATACGAAGATGTCCACG (SEQ ID NO: 15)                        |
|                       | FIP         | CGTACGCGCAACAGT-GGCTTGATGTGGCTCAG (SEQ ID NO: 16)           |
|                       | BIP         | CGTTCATGTGGTCATTCA-TAGAAGCGGTCTGGTCAG (SEQ ID NO: 17)       |
|                       | LF          | GAAAGAAGCAATGAAGTAGC (SEQ ID NO: 18)                        |
|                       | LB          | GAACTAACATTCTTCTCAACG (SEQ ID NO: 19)                       |
| Primer set d (S gene) | F3          | TACTTGGACAATCAAAAGAGTT (SEQ ID NO: 21)                      |
|                       | B3          | GTCTGTAGTAATGATTGTGGTTC (SEQ ID NO: 22)                     |
|                       | B4          | TGCTGTAGTAATGATTGTGG (SEQ ID NO: 23)                        |
|                       | FIP         | CTTTTCTTGTGCAGGGACA-CTATCATCTTATGTCCTTCCCTC (SEQ ID NO: 24) |

TABLE 2-continued

| Primer name | Base sequence                                              |
|-------------|------------------------------------------------------------|
| BIP         | TCACAACTGCTCCTGCC-<br>TGAACAAAGACACCTTCACG (SEQ ID NO: 25) |
| LF          | ACATGCAAGAAGACTACACCA (SEQ ID NO: 26)                      |
| LB          | GATGGAAAAGCACACTTTCCT (SEQ ID NO: 27)                      |

Furthermore, the present invention is a kit comprising the above-described primer set c or d and a fluorescence labeling probe. A probe for the primer set c is a probe consisting of SEQ ID NO: 20, and a probe for the primer set d is a probe consisting of SEQ ID NO: 28. Cytosine bases at the 3' terminals of these probes are fluorescently labeled with BODIPY (registered trademark) FL dye. These probes are Q-probes. That is, when the probes are bound to target nucleic acids, fluorescent dyes and guanine bases of the target nucleic acids are in close proximity, and the fluorescence intensity of the fluorescent dyes is reduced by a quenching action of the guanine base.

TABLE 3

|                        | Base sequence                                                |
|------------------------|--------------------------------------------------------------|
| Probe for primer set c | ATCCAGAACTAACATTCTTCTCAACGTGCCACTCC-bodipyFL (SEQ ID NO: 20) |
| Probe for primer set d | GCAAGAAGACTACACCATGAGGTGCTGAC-bodipyFL (SEQ ID NO: 28)       |

A method for detecting SARS-CoV-2 virus of the present invention is a method for performing an amplification reaction through a LAMP method using the primer set of the present invention.

Enzymes used in nucleic acid synthesis are not particularly limited as long as these are template-dependent nucleic acid synthase having a strand substitution activity. Examples of such enzymes include Bst DNA polymerase (large fragment), Bca(exo-) DNA polymerase, and Klenow fragment of *Escherichia coli* DNA polymerase I, and preferably include Bst DNA polymerase (large fragment).

Reverse transcriptases used in an RT-LAMP method are not particularly limited as long as these are enzymes having an activity of synthesizing DNA using RNA as a template. Examples of such enzymes include reverse transcriptases derived from AMV, Cloned AMV, or MMLV, Superscript II, ReverTra Ace, and Thermoscript, and preferably include reverse transcriptases derived from AMV or Cloned AMV. In addition, when an enzyme, such as Bca DNA polymerase, having both a reverse transcriptase activity and a DNA polymerase activity is used, the RT-LAMP reaction can be performed with one enzyme.

Enzymes or reverse transcriptases used in nucleic acid synthesis may be purified from viruses, bacteria, or the like or may be produced by a gene recombination technique. In addition, these enzymes may be modified through, for example, fragmentation or substitution with an amino acid.

A method for testing for COVID-19 of the present invention is a method for detecting amplification of a target nucleic acid region of SARS-CoV-2 virus using the primer set of the present invention to test for presence or absence of infection with SARS-CoV-2 virus. Well-known techniques

can be applied to detection of nucleic acid amplification products after a LAMP reaction. For example, the detection can be easily performed through a method in which a labeled oligonucleotide that specifically recognizes an amplified sequence or a fluorescent intercalator method (JP 2001-242169) is used or through subjecting a reaction solution after the completion of a reaction directly to agarose gel electrophoresis. In agarose gel electrophoresis, a large number of bands with different base lengths are detected in a ladder shape in LAMP amplification products. In addition, since insoluble magnesium pyrophosphate which is produced through a synthetic reaction of nucleic acids is produced in a LAMP method, a reaction solution is cloudy to the extent that it can be visually observed. A nucleic acid amplification reaction can also be detected by optically measuring such cloudiness (WO 01/83817). Furthermore, calcein, a metal indicator, can also be added to a reaction solution to detect a change in concentration of metal ions as a change in fluorescence with the progress of an amplification reaction (JP 2004-283161).

In a method for diagnosing COVID-19, the primer set of the present invention can be pre-packaged into a kit together with various reagents necessary for detecting nucleic acid amplification. Specifically, various oligonucleotides required as the primers or the loop primers of the present invention, 4 kinds of dNTPs as substrates for nucleic acid synthesis, DNA polymerases for performing nucleic acid synthesis, reverse transcriptases, buffer solutions or salts that provide suitable conditions for an enzymatic reaction, protective agents that stabilize enzymes or templates, and reagents required for detecting reaction products as necessary are provided as a kit.

## EXAMPLES

Hereinafter, the present invention will be described specifically with reference to examples. However, the present invention is not limited by these examples.

### Example 1. Confirmation of Detection Sensitivity of Primer Sets a and b

The detection sensitivity of a LAMP method was confirmed.

#### 1. Preparation of Sample and Reagent

##### 1) Sample (Transcribed RNA)

An RNA template was prepared by integrating a plasmid with cDNA prepared through RT-PCR from a SARS-CoV-2 gene and transcribing and purifying RNA from the plasmid DNA. Script Max (registered trademark) Thermo T7 Transcription Kit (manufactured by TOYOBO Co., LTD., Code Number: TSK-101) was used for the transcription, and RNeasy (registered trademark) Mini Kit (manufactured by QIAGEN N.V., Catalog Number: No. 74104) was used for RNA purification. Diluted solutions of 10 copies to 10<sup>3</sup> copies per microliter were prepared from the purified RNA and used as sample solutions. In addition, a yeast RNA solution was used as a sample solution of 0 copies (negative control).

##### 2) Concentration and Composition of Reagent Used in LAMP Method

25  $\mu$ L of a LAMP reaction reagent having following composition was prepared in a 0.2 mL reagent tube. As a primer, the primer set a or b shown in Table 1 was used.

20 mM Tricine (pH 8.6),  
30 mM KCl,  
8 mM MgSO<sub>4</sub>,

1.4 mM dNTPs,  
0.5% Tween 20,  
1.6 mM DTT,  
1.6  $\mu$ M FIP Primer and BIP Primer,  
0.2  $\mu$ M F3 Primer and B3 Primer,  
0.8  $\mu$ M Loop Primer F (LF) and Loop Primer B (LB),  
AMV Reverse Transcriptase 1.0 U (20 U/ $\mu$ L, Manufactured by Roche Molecular Systems, Inc.),  
Bst DNA polymerase 22.8 U (Manufactured by New England Biolabs),  
RNase Inhibitor (40 U/ $\mu$ L, Manufactured by Promega) 1  $\mu$ L, and  
RNA template (100 copies) 5  $\mu$ L.

## 2. Reaction by Nucleic Acid Amplification Method

### 1) Reaction by LAMP Method

1  $\mu$ L of each of sample solutions containing 0 or 10 to  $10^3$  copies of the target sequence were added to the LAMP reaction reagent prepared in 2) of 1 to obtain 25  $\mu$ L of each final reaction solution. A real-time turbidity measurement device LoopampEXIA (registered trademark) was used for measurement, and the reaction conditions of 63° C. and 60 minutes were set.

### 3. Measurement Results

Measurement results for each of the primer sets are shown in Table 4. It was found that the number of copies that can be detected was up to 10 in the primer sets a with the N gene as a detection target and up to 50 in the primer sets b with the RdRP gene as a detection target.

TABLE 4

| N Gene Use of transcribed RNA<br>Detection time: Minute    |                 |                 |                  |      |
|------------------------------------------------------------|-----------------|-----------------|------------------|------|
| Primer set a                                               |                 |                 |                  |      |
| Number of<br>copies                                        | 1               | 2               | Average<br>value | S.D. |
| 1,000                                                      | 16.4            | 16.2            | 16.3             | 0.14 |
| 500                                                        | 16.5            | 16.7            | 16.6             | 0.14 |
| 200                                                        | 18.0            | 17.7            | 17.9             | 0.21 |
| 100                                                        | 18.4            | 17.6            | 18.0             | 0.57 |
| 50                                                         | 23.3            | 20.2            | 21.8             | 2.19 |
| 20                                                         | 20.1            | 23.0            | 21.6             | 2.05 |
| 10                                                         | 25.0            | 37.3            | 31.2             | 8.70 |
| Negative<br>control                                        | Not<br>detected | Not<br>detected |                  |      |
| RdRP Gene Use of transcribed RNA<br>Detection time: Minute |                 |                 |                  |      |
| Primer set b                                               |                 |                 |                  |      |
| Number of<br>copies                                        | 1               | 2               | Average<br>value | S.D. |
| 1,000                                                      | 17.6            | 17.6            | 17.6             | 0.00 |
| 500                                                        | 19.2            | 18.3            | 18.8             | 0.64 |
| 200                                                        | 22.8            | 17.7            | 20.3             | 3.61 |
| 100                                                        | 26.5            | 21.6            | 24.1             | 3.46 |
| 50                                                         | 24.6            | 26.0            | 25.3             | 0.99 |
| 20                                                         | Not<br>detected | Not<br>detected |                  |      |
| 10                                                         | 28.2            | Not<br>detected | 28.2             |      |
| Negative<br>control                                        | Not<br>detected | Not<br>detected |                  |      |

### Example 2. Detection Sensitivity when Genomic RNA was Used as Template

Viral genomic RNA dispensed from the National Institute of Infectious Disease was used to examine the detection

sensitivity. Concurrently, PCR was also carried out for comparison of the detection sensitivity therebetween.

## 1. Preparation of Sample and Reagent

### 1) Sample (Genomic RNA)

Diluted solutions of 50 copies to  $1.6 \times 10^3$  copies of genomic RNA of SARS-CoV-2 dispensed from the National Institute of Infectious Disease per microliter were prepared used as sample solutions.

### 2) Concentration and Composition of Reagent Used in LAMP Method

A reaction reagent was prepared in the same manner as in Example 1. The primer set a with the N gene as a detection target was used as a primer set.

### 3) Concentration and Composition of Reagent Used in PCR

An amplification reagent and a primer with the N gene as a detection target were prepared in accordance with “Pathogen Detection Manual 2019-nCoV Ver.2.7” of the National Institute of Infectious Disease.

## 2. Reaction by Nucleic Acid Amplification Method

### 1) Reaction by LAMP Method

An amplification reaction was carried out with a real-time turbidity measurement device LoopampEXIA (registered trademark) in the same manner as in Example 1 at 63° C. for 60 minutes.

### 2) Reaction by PCR

An RT-PCR method in which TaqMan probe was used was carried out in accordance with “Pathogen Detection Manual 2019-nCoV Ver.2.7” of the National Institute of Infectious Disease.

### 3. Measurement Results

The measurement results in each method are shown in Table 5. Although genomic RNA up to 50 copies was able to be detected in both amplification methods, the detection in the LAMP method was able to be performed in a shorter period of time than PCR.

TABLE 5

| Measurement results for Genomic RNA<br>Detection time: Minute |                 |                 |                 |                  |      |
|---------------------------------------------------------------|-----------------|-----------------|-----------------|------------------|------|
| LAMP Method                                                   |                 |                 |                 |                  |      |
| Number of<br>copies                                           | 1               | 2               | 3               | Average<br>value | S.D. |
| 1,600                                                         | 20.3            | 19.7            | 19.3            | 19.8             | 0.5  |
| 800                                                           | 20.0            | 19.6            | 20.2            | 19.9             | 0.3  |
| 400                                                           | 21.2            | 21.9            | 21.6            | 21.6             | 0.4  |
| 200                                                           | 23.2            | 25.0            | 22.7            | 23.6             | 1.2  |
| 100                                                           | 24.7            | 26.6            | 23.1            | 24.8             | 1.8  |
| 50                                                            | 26.3            | 26.6            | 35.9            | 29.6             | 5.5  |
| Negative<br>control                                           | Not<br>detected | Not<br>detected | Not<br>detected |                  |      |
| Detection time: Minute                                        |                 |                 |                 |                  |      |
| PCR                                                           |                 |                 |                 |                  |      |
| Number of<br>copies                                           | 1               | 2               | 3               | Average<br>value | S.D. |
| 1,600                                                         | 29.5            | 29.6            | 29.6            | 29.6             | 0.1  |
| 800                                                           | 30.5            | 30.7            | 30.6            | 30.6             | 0.1  |
| 400                                                           | 31.5            | 31.5            | 31.5            | 31.5             | 0.0  |
| 200                                                           | 32.7            | 32.5            | 32.6            | 32.6             | 0.1  |
| 100                                                           | 33.2            | 33.5            | 34.2            | 33.6             | 0.5  |
| 50                                                            | 34.6            | 34.7            | 35.0            | 34.8             | 0.2  |
| Negative<br>control                                           | Not<br>detected | Not<br>detected | Not<br>detected |                  |      |

## 11

Example 3. Confirmation of Detection Sensitivity  
of Primer Sets c and d

The detection sensitivity of a LAMP method was confirmed.

## 1. Preparation of Sample and Reagent

## 1) Sample (Transcribed RNA)

Artificial genes in which sequences of an M region and an S region were respectively incorporated were synthesized, and transcribed RNA copies thereof were used as templates. Synthesis of the artificial genes was outsourced to Eurofins. The sequences of the artificial genes are represented by SEQ ID NO: 29 (artificial gene for detecting M region), and SEQ ID NO: 30 (artificial gene for detecting S region).

## 2) Concentration and Composition of Reagent Used in LAMP Method

A LAMP reaction reagent having the following composition was prepared.

LAMP Reaction reagent:

20 mM Tricine (pH 8.6),

30 mM KCl,

0.1% Tween 20,

1.4 mM dNTPs,

8 mM MgSO<sub>4</sub>,

1.6 mM DTT,

PPase 20 mU

Bst DNA polymerase 25 U,

RNase Inhibitor 1.0 U, and

AMV Reverse Transcriptase 1.0 U.

A primer set c or d, a fluorescence labeling probe corresponding to the primer set, and SYTO™ 63 Red Fluorescent Nucleic Acid Stain were added to the reaction reagent, a master mix (MM) prepared with a purified water (DW) to have a concentration of 15 µL/test was used, and a total of 25 µL including 5 µL of DW, 5 µL of a template, and the master mix was used in one reaction. The reaction reagent was prepared on ice. Master mix (amount per reaction):

LAMP Reaction reagent 12.5 µL

5 µM SYTO 0.5 µL

10 µM Probe 0.1 µL

Primer set Appropriate amount

DW Total of 15 µL.

Amount per reaction:

Master mix 15 µL

DW 5 µL

RNA template 5 µL.

## 2. Reaction by Nucleic Acid Amplification Method

## 1) Reaction by LAMP Method

A real-time quantitative PCR system LightCycler (registered trademark) 96 (manufactured by Roche Molecular Systems, Inc.) was used for measurement, and evaluation was performed at a reaction temperature of 63° C.

## 3. Measurement Results

Measurement results for each of the primer sets are shown in Table 6.

TABLE 6

| Detection time: Minute |              |      |      |      |         |
|------------------------|--------------|------|------|------|---------|
| Number of              | Primer set c |      |      |      |         |
| copies                 | 1            | 2    | 3    | 4    | Average |
| 1,000                  | 6.4          | 6.5  | —    | —    | 6.5     |
| 50                     | 12.6         | 9.6  | 10.3 | 8.9  | 10.4    |
| 25                     | 10.2         | 10.8 | 9.0  | 13.1 | 10.8    |
| 12.5                   | 9.8          | 12.6 | 19.3 | 11.7 | 13.4    |

## 12

TABLE 6-continued

| Number of<br>copies | Negative<br>control    | Not<br>detected | Not<br>detected | —               | —               | —       |
|---------------------|------------------------|-----------------|-----------------|-----------------|-----------------|---------|
|                     | Detection time: Minute |                 |                 |                 |                 |         |
| Primer set d        |                        |                 |                 |                 |                 |         |
|                     | 1                      | 2               | 3               | 4               | 5               | Average |
| 1,000               | 5.9                    | 5.4             | 5.7             | 5.8             | 6.0             | 5.8     |
| 100                 | 6.9                    | 7.8             | 6.5             | 8.2             | 7.4             | 7.4     |
| 50                  | 7.4                    | 8.2             | 8.9             | 8.1             | 8.1             | 8.1     |
| 25                  | 9.8                    | 10.0            | 12.7            | 8.3             | 8.8             | 9.9     |
| 12.5                | Not<br>detected        | 12.2            | Not<br>detected | 10.0            | 18.1            | 13.4    |
| Negative<br>control | Not<br>detected        | Not<br>detected | Not<br>detected | Not<br>detected | Not<br>detected | —       |

The minimum detection sensitivity that could be detected within 15 minutes was 25 copies/test in any system in which any primer set was used.

Example 2. Confirmation of Cross-Reactivity with  
SARS Virus

At GGenome (<https://gggenome.dbcls.jp/ja/>), cross-reactivity with 38 kinds of respiratory system-related bacteria and pathogenic microorganisms (Human coronavirus 229E, Human coronavirus OC43, Human coronavirus HKU1, Human coronavirus NL63, SARS-coronavirus, SARS-coronavirus-2, MERS-coronavirus, Adenovirus, Human Metapneumovirus, Parainfluenza virus 1-4, Influenza A, Influenza B, Enterovirus, Respiratory syncytial virus, Rhinovirus, *Chlamydia pneumoniae*, *Haemophilus influenzae*, *Legionella pneumophila*, *Mycobacterium tuberculosis*, *Streptococcus pneumoniae*, *Streptococcus pyogenes*, *Bordetella pertussis*, *Mycoplasma pneumoniae*, *Pneumocystis jirovecii* (PJP), Influenza C, Parechovirus, *Candida albicans*, *Corynebacterium diphtheriae*, *Legionella non-pneumophila*, *Bacillus anthracis* (Anthrax), *Moraxella cararrhalis*, *Neisseria elongate* and *meningitidis*, *Pseudomonas aeruginosa*, *Staphylococcus epidermis*, *Staphylococcus salivarius*, *Lep- tospirosis*, *Chlamydia psittaci*, *Coxiella burneti* (Q-Fever), *Streptococcus aureus*) was confirmed in silico for each of the M region and the S region.

As a result, in only an M region of SARS-Coronavirus (SARS virus), a mismatch in a probe design region was within 5 bases and each of mismatches in F1, F2, B1, and B2 primer design regions was within 3 bases, and these became positions for a LAMP reaction. Therefore, the cross-reactivity with SARS virus was confirmed.

Transcribed RNA of an artificial gene with which a sequence of SARS virus (GenBank NC004718.3) having high homology for each region was integrated was used as a template of the SARS virus (synthesis request destination: Eurofins Scientific). The amount of template was 1.0×10<sup>7</sup> copies/test.

Measurement results for each of the primer sets are shown in Table 7.

## 13

TABLE 7

| Detection time: Minute             |              |              |              |              |         |
|------------------------------------|--------------|--------------|--------------|--------------|---------|
| Template                           | Primer set c |              |              |              | Average |
|                                    | 1            | 2            | 3            | 4            |         |
| 1 × 10 <sup>7</sup> Copies of SARS | Not detected | Not detected | Not detected | Not detected | —       |
| 1,000 Copies of SARS-CoV-2         | 5.6          | 5.6          | —            | —            | 5.6     |
| Negative control                   | Not detected | Not detected | —            | —            | —       |
| Detection time: Minute             |              |              |              |              |         |
| Template                           | Primer set d |              |              |              | Average |
|                                    | 1            | 2            | 3            | 4            |         |
| 1 × 10 <sup>7</sup> Copies of SARS | Not detected | Not detected | Not detected | Not detected | —       |
| 1,000 Copies of SARS-CoV-2         | 5.7          | 5.7          | —            | —            | 5.7     |
| Negative control                   | Not detected | Not detected | —            | —            | —       |

No cross-reaction with SARS virus was observed in both detection systems. In addition, assuming a case of co-infection with SARS-CoV-2 and SARS virus, tests in which 1.0×10<sup>6</sup> copies/test of transcribed RNA of SARS virus was mixed with 25 or 1,000 copies/test of transcribed RNA of SARS-CoV-2 were performed.

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Measurement results for each of the primer sets are shown in Table 8.

TABLE 8

| Detection time: Minute             |              |              |              |      |         |
|------------------------------------|--------------|--------------|--------------|------|---------|
| Template                           | Primer set c |              |              |      | Average |
|                                    | 1            | 2            | 3            | 4    |         |
| 1,000 Copies of SARS-CoV-2         | 5.2          | 5.2          | 5.1          | 5.2  | 5.2     |
| 25 Copies of SARS-CoV-2            | 5.2          | 5.2          | 5.3          | 5.2  | 5.2     |
| 1 × 10 <sup>6</sup> Copies of SARS | 6.7          | 7.9          | 7.2          | 7.3  | 7.3     |
| No SARS                            | 6.9          | 7.5          | 6.9          | 7.1  | 7.1     |
| Negative control                   | Not detected | Not detected | Not detected | —    | —       |
| Detection time: Minute             |              |              |              |      |         |
| Template                           | Primer set d |              |              |      | Average |
|                                    | 1            | 2            | 3            | 4    |         |
| 1,000 Copies of SARS-CoV-2         | 5.4          | 5.5          | 5.4          | 5.4  | 5.4     |
| 25 Copies of SARS-CoV-2            | 5.5          | 5.5          | 5.6          | 5.5  | 5.5     |
| 1 × 10 <sup>6</sup> Copies of SARS | 7.8          | 9.9          | 10.0         | 9.2  | 9.2     |
| No SARS                            | 11.3         | 10.3         | 12.3         | 11.3 | 11.3    |
| Negative control                   | Not detected | Not detected | Not detected | —    | —       |

SARS-CoV-2 could also be detected in the mixing test.

## SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 30

<210> SEQ ID NO 1

<211> LENGTH: 29903

<212> TYPE: DNA

<213> ORGANISM: SARS-CoV-2

<400> SEQUENCE: 1

```

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cacgcagtat aattaataac taattactgt cgttgacagg acacgagtaa ctcgctctatc      180
ttctgcaggc tgcttacggg ttctgcccgtg ttgcagccga tcatcagcac atctaggttt      240
cgtccgggtg tgaccgaaaag gtaagatgga gagccttgtc cctggtttca acgagaaaac      300
acacgtccaa ctcagtttgc ctgttttaca ggttcgcgac gtgctcgtac gtggctttgg      360
agactccgtg gaggagggtc tatcagaggg acgtcaacat cttaaagatg gcaactgtgg      420
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cgaaggcatt cagtacggtc gtagtggtga gacacttggt gtccttgtcc ctcatgtggg      600
cgaaatacca gtggcttacc gcaaggttct tcttcgtaag aacggtaata aaggagctgg      660
tggccatagt tacggcgccg atctaaagtc atttgactta ggcgacgagc ttggcactga      720
tccttatgaa gatttttcaag aaaactggaa cactaaacat agcagtgggt ttaccctgta      780

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|             |             |             |            |             |             |      |
|-------------|-------------|-------------|------------|-------------|-------------|------|
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| ccctgatggc  | tacctcttg   | agtgcattaa  | agaccttcta | gcacgtgctg  | gtaaagcttc  | 900  |
| atgcactttg  | tccgaacaac  | tggaactttat | tgacactaag | aggggtgtat  | actgtgccg   | 960  |
| tgaacatgag  | catgaaattg  | cttggtacac  | ggaacgttct | gaaaagagct  | atgaattgca  | 1020 |
| gacacctttt  | gaaattaaat  | tggaacagaa  | atgtgacacc | ttcaatgggg  | aatgtccaaa  | 1080 |
| ttttgtattt  | cccttaaatt  | ccataatcaa  | gactattcaa | ccaagggttg  | aaaagaaaaa  | 1140 |
| gcttgatggc  | tttatgggta  | gaattcgatc  | tgtctatcca | gttgcgtcac  | caaatgaatg  | 1200 |
| caaccaaagt  | tgcctttcaa  | ctctcatgaa  | gtgtgatcat | tgtggtgaaa  | cttcatggca  | 1260 |
| gacggggcat  | tttgtaaag   | ccacttgcca  | attttggggc | actgagaatt  | tgactaaaga  | 1320 |
| aggtgccact  | acttgtgggt  | acttacccca  | aaatgctgtt | gttaaaattt  | attgtccagc  | 1380 |
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| ttatgttggt  | tgccataaca  | agtgtgcta   | ttgggttcca | cgtgctagcg  | ctaacatagg  | 1560 |
| ttgtaacct   | acagggtgtg  | ttggagaagg  | ttccgaaggt | cttaatgaca  | accttcttga  | 1620 |
| aatactccaa  | aaagagaaag  | tcaacatcaa  | tattgttggt | gactttaaac  | ttaatgaaga  | 1680 |
| gatcgccatt  | attttggcat  | ctttttctgc  | ttccacaagt | gcttttgtgg  | aaactgtgaa  | 1740 |
| aggtttggat  | tataaagcat  | tcaacaaaat  | tgttgaatcc | tgtggtaatt  | ttaaagttac  | 1800 |
| aaaaggaaaa  | gctaaaaaag  | gtgcctggaa  | tattggtgaa | cagaaatcaa  | tactgagtcc  | 1860 |
| tctttatgca  | tttgcatcag  | aggtgctcg   | tgtgtacga  | tcaattttct  | cccgcaactc  | 1920 |
| tgaaactgct  | caaaattctg  | tgcgtgtttt  | acagaaggcc | gctataacaa  | tactagatgg  | 1980 |
| aatttcacag  | tattcactga  | gactcattga  | tgcctatgat | ttcacatctg  | atttggctac  | 2040 |
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| agagaagttt  | aagggaagtg  | tagagtttct  | tagagacggt | tgggaaattg  | ttaaatttat  | 2220 |
| ctcaacctgt  | gcttgtgaaa  | ttgtcgggtg  | acaaattgtc | acctgtgcaa  | aggaaattaa  | 2280 |
| ggagagtgtt  | cagacattct  | ttaagcttgt  | aaataaattt | ttggctttgt  | gtgctgactc  | 2340 |
| tatcattatt  | gggtggagcta | aacttaaaagc | cttgaattta | gggtgaaacat | ttgtcacgca  | 2400 |
| ctcaaaggga  | ttgtacagaa  | agtgtgttaa  | atccagagaa | gaaactggcc  | tactcatgcc  | 2460 |
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| agctgttgaa  | gctccattgg  | ttggtacacc  | agtttgtatt | aacgggctta  | tgttgctcga  | 2640 |
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| ctgtgttgtg  | gcagatgctg  | tcataaaaa   | tttgcaacca | gtatctgaat  | tacttacacc  | 2940 |
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| tgagtttaaa  | ttggcttcac  | atatgtattg  | ttctttctac | cctccagatg  | aggatgaaga  | 3060 |
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|                                                                    |      |
|--------------------------------------------------------------------|------|
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| taacttttagc | ttgtttttgtg | cttgctgctg  | tttacagaat  | aaattggatc  | acgggtggaa  | 26760 |
| ttgctatcgc  | aatggcttgt  | cttgtaggct  | tgatgtggct  | cagctacttc  | attgcttctt  | 26820 |
| tcagactgtt  | tgcgcgtacg  | cgttccatgt  | ggtcattcaa  | tccagaaact  | aacattcttc  | 26880 |

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|            |            |             |            |             |            |       |
|------------|------------|-------------|------------|-------------|------------|-------|
| tcaacgtgcc | actccatggc | actattctga  | ccagaccgct | tctagaaagt  | gaactcgtaa | 26940 |
| tcggagctgt | gacccctcgt | ggacatcttc  | gtattgctgg | acaccatcta  | ggacgctgtg | 27000 |
| acatcaagga | cctgcctaaa | gaaatcactg  | ttgctacatc | acgaacgctt  | tcttattaca | 27060 |
| aattgggagc | ttcgacagct | gtagcaggtg  | actcaggttt | tgetgcatac  | agtcgctaca | 27120 |
| ggattggcaa | ctataaatta | aacacagacc  | attccagtag | cagtgcacat  | attgctttgc | 27180 |
| ttgtacagta | agtgacaaca | gatgtttcat  | ctcgttgact | ttcaggttac  | tatagcagag | 27240 |
| atattactaa | ttattatgag | gactttttaa  | gtttccattt | ggaatcttga  | ttacatcata | 27300 |
| aaacctcata | ttaaaaattt | atctaagtca  | ctaactgaga | ataaatattc  | tcaattagat | 27360 |
| gaagagcaac | caatggagat | tgattaaacg  | aacatgaaaa | ttattctttt  | cttggcactg | 27420 |
| ataacactcg | ctacttgtga | gctttatcac  | taccaagagt | gtgttagagg  | tacaacagta | 27480 |
| cttttaaaag | aaccttgctc | ttctggaaca  | tacgagggca | attcaccatt  | tcacccctca | 27540 |
| gctgataaca | aatttgactc | gacttgcttt  | agcactcaat | ttgcttttgc  | ttgtcctgac | 27600 |
| ggcgtaaaac | acgtctatca | gttacgtgcc  | agatcagttt | cacctaaact  | gttcacacga | 27660 |
| caagaggaag | ttcaagaact | ttactctcca  | atttttctta | ttgttgcggc  | aatagtgttt | 27720 |
| ataacacttt | gcttcacact | caaaagaaag  | acagaatgat | tgaactttca  | ttaattgact | 27780 |
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| gggagccttg | aatacaccaa | aagatcacat  | tggcaccggc | aatcctgcta  | acaatgctgc | 28740 |
| aatcgtgcta | caacttcctc | aaggaacaa   | attgccaaaa | ggcttctacg  | cagaaggag  | 28800 |
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| tgctgctctt | gctttgctgc | tgcttgacag  | attgaaccag | cttgagagca  | aaatgtctgg | 28980 |
| taaaggccaa | caacaacaag | gccaaactgt  | cactaagaaa | tctgctgctg  | aggtctctaa | 29040 |
| gaagcctcgg | caaaaacgta | ctgccactaa  | agcatacaat | gtaacacaag  | ctttcggcag | 29100 |
| acgtggtcca | gaacaaaccc | aaggaaattt  | tggggaccag | gaactaatca  | gacaaggaac | 29160 |
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aatgtcgcgc attggcatgg aagtcacacc ttcgggaacg tgggtgacct acacaggtgc 29280
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ggggaggactt gaaagagcca ccacattttc accgaggcca cgcgaggtag gatcgagtgt 29760
acagtgaaca atgctaggga gagctgcta tatggaagag ccctaattgtg taaaattaat 29820
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<223> OTHER INFORMATION: F3 primer

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<210> SEQ ID NO 3
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: B3 primer

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<400> SEQUENCE: 3

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<210> SEQ ID NO 4
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FIP primer

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<400> SEQUENCE: 4

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<210> SEQ ID NO 5
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: BIP primer

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: LF primer

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<220> FEATURE:  
<223> OTHER INFORMATION: LB primer

<400> SEQUENCE: 7  
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<210> SEQ ID NO 8  
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<223> OTHER INFORMATION: F3 primer

<400> SEQUENCE: 8  
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<210> SEQ ID NO 9  
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<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: B3 primer

<400> SEQUENCE: 9  
acttatcggc aattttgtta cc 22

<210> SEQ ID NO 10  
<211> LENGTH: 46  
<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: FIP primer

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<212> TYPE: DNA  
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<223> OTHER INFORMATION: BIP primer

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<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: LF primer

<400> SEQUENCE: 12  
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<210> SEQ ID NO 13

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<211> LENGTH: 20  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: LB primer

<400> SEQUENCE: 13

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 <212> TYPE: DNA  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: F3 primer

<400> SEQUENCE: 14

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 <220> FEATURE:  
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 <220> FEATURE:  
 <223> OTHER INFORMATION: FIP primer

<400> SEQUENCE: 16

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 <220> FEATURE:  
 <223> OTHER INFORMATION: BIP primer

<400> SEQUENCE: 17

cgttccatgt ggcatcatc agaagcggc ttggtcag 37

<210> SEQ ID NO 18  
 <211> LENGTH: 20  
 <212> TYPE: DNA  
 <213> ORGANISM: Artificial Sequence  
 <220> FEATURE:  
 <223> OTHER INFORMATION: LF primer

<400> SEQUENCE: 18

gaaagaagca atgaagtagc 20

<210> SEQ ID NO 19  
 <211> LENGTH: 22  
 <212> TYPE: DNA  
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 <220> FEATURE:  
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<400> SEQUENCE: 19

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gaaactaaca ttcttctcaa cg 22

<210> SEQ ID NO 20  
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<223> OTHER INFORMATION: Probe  
<220> FEATURE:  
<221> NAME/KEY: misc\_feature  
<222> LOCATION: (36)..(36)  
<220> FEATURE:  
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<222> LOCATION: (36)..(36)  
<223> OTHER INFORMATION: bodipy FL

<400> SEQUENCE: 20

atccagaaac taacattctt ctcaacgtgc cactcc 36

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<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: F3 primer

<400> SEQUENCE: 21

tacttggaaca atcaaaaaga gtt 23

<210> SEQ ID NO 22  
<211> LENGTH: 24  
<212> TYPE: DNA  
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<220> FEATURE:  
<223> OTHER INFORMATION: B3 primer

<400> SEQUENCE: 22

gtctgtagta atgatttggt gttc 24

<210> SEQ ID NO 23  
<211> LENGTH: 22  
<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: B4 primer

<400> SEQUENCE: 23

tgtctgtagt aatgatttgt gg 22

<210> SEQ ID NO 24  
<211> LENGTH: 42  
<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: FIP primer

<400> SEQUENCE: 24

cttttcttgt gcaggacac tatcatetta tgccttccc tc 42

<210> SEQ ID NO 25  
<211> LENGTH: 38  
<212> TYPE: DNA  
<213> ORGANISM: Artificial Sequence  
<220> FEATURE:  
<223> OTHER INFORMATION: BIP primer

<400> SEQUENCE: 25



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|                                                                                                                                                                                                                                                                                                      |     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| tcacaactgc tctgcctga aacaaagaca ccttcacg                                                                                                                                                                                                                                                             | 38  |
| <210> SEQ ID NO 26<br><211> LENGTH: 21<br><212> TYPE: DNA<br><213> ORGANISM: Artificial Sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: LF primer<br><br><400> SEQUENCE: 26                                                                                                                   |     |
| acatgcaaga agactacacc a                                                                                                                                                                                                                                                                              | 21  |
| <210> SEQ ID NO 27<br><211> LENGTH: 21<br><212> TYPE: DNA<br><213> ORGANISM: Artificial Sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: LB primer<br><br><400> SEQUENCE: 27                                                                                                                   |     |
| gatggaaaag cacactttcc t                                                                                                                                                                                                                                                                              | 21  |
| <210> SEQ ID NO 28<br><211> LENGTH: 29<br><212> TYPE: DNA<br><213> ORGANISM: Artificial Sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Probe<br><220> FEATURE:<br><221> NAME/KEY: misc_feature<br><222> LOCATION: (29)..(29)<br><223> OTHER INFORMATION: bodipy FL<br><br><400> SEQUENCE: 28 |     |
| gcaagaagac tacaccatga ggtgctgac                                                                                                                                                                                                                                                                      | 29  |
| <210> SEQ ID NO 29<br><211> LENGTH: 377<br><212> TYPE: DNA<br><213> ORGANISM: Artificial Sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Artificial Sequence for detecting SARS-CoV-2 M region<br><br><400> SEQUENCE: 29                                                                      |     |
| gtaatacgac tcaactatagg gatggccagt aactttagct tgttttgtgc ttgctgctgt                                                                                                                                                                                                                                   | 60  |
| ttacagaata aattggatca ccggtggaat tgctatcgca atggcttgtc ttgtaggctt                                                                                                                                                                                                                                    | 120 |
| gatgtggctc agctacttca ttgcttcttt cagactgttt gcgcgtacgc gttccatgtg                                                                                                                                                                                                                                    | 180 |
| gtcattcaat ccagaaacta acattcttct caacgtgcc aacctatggca ctattctgac                                                                                                                                                                                                                                    | 240 |
| cagaccgctt ctgaaaagtg aactcgtaat cggagctgtg atccttcgtg gacatcttcg                                                                                                                                                                                                                                    | 300 |
| tattgctgga caccatctag gacgctgtga catcaaggac ctgcctaaag aaatcactgt                                                                                                                                                                                                                                    | 360 |
| tgctacatca cgaacgc                                                                                                                                                                                                                                                                                   | 377 |
| <210> SEQ ID NO 30<br><211> LENGTH: 1593<br><212> TYPE: DNA<br><213> ORGANISM: Artificial Sequence<br><220> FEATURE:<br><223> OTHER INFORMATION: Artificial Sequence for detecting SARS-CoV-2 S2 region<br><br><400> SEQUENCE: 30                                                                    |     |

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|                                                                    |      |
|--------------------------------------------------------------------|------|
| gtaatacgac tcaactatagg gaatcttttg ttgcaatatg gcagtttttg tacacaatta | 60   |
| aacccgtgctt taactggaat agctgttgaa caagacaaaa acaccaaga agtttttgca  | 120  |
| caagtcaaac aaatttatac aacaccacca attaaagatt ttggtggttt taatttttca  | 180  |
| caaatattac cagatccatc aaaaccaagc aagaggatcat ttattgaaga tctacttttc | 240  |
| aacaaagtga cacttgcaga tgctggcttc atcaacaat atggtgattg ccttgggtgat  | 300  |
| attgctgcta gagacctcat ttgtgcacaa aagtttaacg gccttactgt ttgcccact   | 360  |
| ttgctcacag atgaaatgat tgctcaatac acttctgcac tggttagcggg tacaatcact | 420  |
| tctggttgga cctttggtgc aggtgctgca ttacaaatac catttgctat gcaaatggct  | 480  |
| tataggttta atggtattgg agttacacag aatgttctct atgagaacca aaaattgatt  | 540  |
| gccaaaccaat ttaatagtgc tattggcaaa attcaagact cactttcttc cacagcaagt | 600  |
| gcacttgga aacttcaaga tgtggtcaac caaatgcac aagctttaa cacgcttggt     | 660  |
| aaacaactta gctccaattt tgggtgaatt tcaagtgttt taaatgatat cctttcacgt  | 720  |
| cttgacaaag ttgaggctga agtgcaaat gataggttga tcacaggcag acttcaaagt   | 780  |
| ttgcagacat atgtgactca acaattaatt agagctgcag aaatcagagc ttctgcta    | 840  |
| cttgctgcta ctaaaatgct agagtgtgta cttggacaat caaaaagagt tgatttttgt  | 900  |
| ggaaagggt atcatcttat gtccttcct cagtcagcac ctcatggtgt agtctcttg     | 960  |
| catgtgactt atgtccctgc acaagaaaag aacttcacaa ctgctcctgc catttgctcat | 1020 |
| gatggaaaag cacactttcc tctgaaggt gtctttgttt caaatggc acactgggtt     | 1080 |
| gtaacacaaa ggaattttta tgaaccacaa atcattacta cagacaacac atttgtgtct  | 1140 |
| ggtaactgtg atgttgtaatt aggaattgtc aacaacacag tttatgatcc ttgcaacct  | 1200 |
| gaattagact cattcaagga ggaggttagat aaatatttta agaatacaca atcaccagat | 1260 |
| gttgatttag gtgacatctc tggcattaat gcttcagttg taaacattca aaagaaatt   | 1320 |
| gaccgcctca atgaggttgc caagaattta aatgaatctc tcatgatct ccaagaactt   | 1380 |
| ggaaagtatg agcagtatat aaaatggcca tggtagattt ggctagggtt tatagctggc  | 1440 |
| ttgattgcca tagtaatggt gacaattatg ctttgtgtga tgaccagttg ctgtagttgt  | 1500 |
| ctcaagggt gttgttcttg tggatcctgc tgcaaatgtg atgaagacga ctctgagcca   | 1560 |
| gtgctcaag gagtcaaat acattacaca taa                                 | 1593 |

The invention claimed is:

1. A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method,

wherein the primer set consists of at least one selected from the group consisting of (c1) and (d1):

(c1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 14, a B3 primer consisting of SEQ ID NO: 15, an FIP primer consisting of SEQ ID NO: 16, and a BIP primer consisting of SEQ ID NO: 17; and

(d1) a primer set consisting of an F3 primer consisting of SEQ ID NO: 21, a B3 primer consisting of SEQ ID NO: 22, a B4 primer consisting of SEQ ID NO: 23, an FIP primer consisting of SEQ ID NO: 24, and a BIP primer consisting of SEQ ID NO: 25.

2. A primer set for detecting SARS-CoV-2 virus through amplification by a LAMP method,

wherein the primer set consists of at least one selected from the group consisting of (c2) and (d2):

(c2) a primer set consisting of an F3 primer consisting of SEQ ID NO: 14, a B3 primer consisting of SEQ ID NO: 15,

an FIP primer consisting of SEQ ID NO: 16, a BIP primer consisting of SEQ ID NO: 17, an LF primer consisting of SEQ ID NO: 18, and an LB primer consisting of SEQ ID NO: 19; and

(d2) a primer set consisting of an F3 primer consisting of SEQ ID NO: 21, a B3 primer consisting of SEQ ID NO: 22, a B4 primer consisting of SEQ ID NO: 23, an FIP primer consisting of SEQ ID NO: 24, a BIP primer consisting of SEQ ID NO: 25, an LF primer consisting of SEQ ID NO: 26, and an LB primer consisting of SEQ ID NO: 27.

3. A kit for detecting SARS-CoV-2 virus through amplification by a LAMP method, comprising:

the primer set of claim 1; and  
a fluorescence labeling probe.

4. The kit of claim 3,

wherein

the fluorescence labeling probe is a probe consisting of SEQ ID NO: 20.

5. The kit of claim 3, which is used for testing for COVID-19.

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6. A method for detecting SARS-CoV-2 virus, comprising:  
 performing an amplification reaction through a LAMP  
 method using the primer set of claim 1.
7. A method for testing for COVID-19, comprising:  
 detecting amplification of a target nucleic acid region of  
 SARS-CoV-2 virus using the primer set of claim 1 to  
 test for presence or absence of infection with SARS-  
 CoV-2 virus.
8. A method for detecting SARS-CoV-2 virus, comprising:  
 performing an amplification reaction through a LAMP  
 method using the kit of claim 3.
9. A method for testing for COVID-19, comprising:  
 detecting amplification of a target nucleic acid region of  
 SARS-CoV-2 virus using the kit of claim 3 to test for  
 presence or absence of infection with SARS-CoV-2  
 virus.
10. A kit for detecting SARS-CoV2 virus through ampli-  
 fication by a LAMP method, comprising:  
 the primer set of claim 2; and  
 a fluorescence labeling probe.

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11. The kit of claim 10,  
 wherein the fluorescence labeling probe is a probe con-  
 sisting of SEQ ID NO: 20.
12. The kit of claim 10, which is used for testing for  
 COVID-19.
13. A method for detecting SARS-CoV-2 virus, compris-  
 ing:  
 performing an amplification reaction through a LAMP  
 method using the primer set of claim 2.
14. A method for testing for COVID-19, comprising:  
 detecting amplification of a target nucleic acid region of  
 SARS-CoV-2 virus using the primer set of claim 2 to  
 test for presence or absence of infection with SARS-  
 CoV-2 virus.
15. A method for detecting SARS-CoV-2 virus, compris-  
 ing:  
 performing an amplification reaction through a LAMP  
 method using the kit of claim 10.
16. A method for testing for COVID-19, comprising:  
 detecting amplification of a target nucleic acid region of  
 SARS-CoV-2 virus using the kit of claim 10 to test for  
 presence or absence of infection with SARS-CoV-2  
 virus.

\* \* \* \* \*